

**AI-BASED PREDICTIVE MAINTENANCE IN INDIAN RAILWAYS:
FALSE ALARMS, SAFETY RISKS, EXPLAINABILITY AND RESPONSIBLE
DECISION-MAKING**

A CASE STUDY

**Submitted for
CS23062 – RESPONSIBLE AI**

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1. ABSTRACT

Railway systems contain many components that need regular inspection and maintenance. Components such as wheels, bearings, brakes, tracks, signaling equipment and other mechanical parts can develop faults because of continuous operation. Traditionally, maintenance is planned at fixed intervals or carried out when a problem is noticed during inspection. With sensor-based monitoring and Artificial Intelligence, predictive maintenance can be used to identify possible equipment failures before they become serious.

In an AI-based predictive maintenance system, information collected from sensors and inspection systems can be analyzed by a machine learning model. The model may estimate whether a component is operating normally or whether it needs further inspection. This can help maintenance teams plan their work and reduce unexpected failures.

However, using AI in railway maintenance is also a Responsible AI problem. A prediction can influence an important safety decision. A false negative, where the system fails to identify a developing fault, could create serious safety risk. A false positive may result in unnecessary inspection, replacement or maintenance, causing additional cost and operational disruption.

This case study examines AI-based predictive maintenance in the Indian Railways context from a Responsible AI perspective. It focuses on safety, reliability, fairness, explainability, accountability, human oversight, security and privacy. The proposed approach treats AI as a decision-support tool rather than allowing it to independently make maintenance decisions.

Explainability methods such as SHAP and feature importance can be used to understand why a model has predicted a possible failure. A confidence-based human review mechanism can also be introduced so that uncertain or high-risk predictions are checked by qualified maintenance personnel before action is taken.

2.DOMAIN

The case study lies at the intersection of Responsible Artificial Intelligence, predictive maintenance, machine learning, Intelligent Transportation Systems, railway safety, Industrial IoT and infrastructure monitoring.

A simplified concept is:

Stage	Purpose
Railway component	Equipment whose condition is being monitored
Sensors / inspection data	Collect operating and condition information
Data preprocessing	Clean and prepare the readings
AI / ML model	Estimate failure risk or equipment condition
Explainability	Show important factors behind a prediction
Engineer review	Check the prediction against practical evidence
Maintenance decision	Decide inspection, repair, replacement or continued monitoring

3. REAL-WORLD PROBLEM

Consider a railway component that has been operating normally for a long period. Its sensor readings may slowly change because of wear, overheating, vibration or another developing fault. A predictive maintenance model may recognize these changes before they become obvious during a normal inspection.

For example, the system may produce an alert such as: “Possible component failure – high risk – engineer inspection recommended.” This is useful only when the alert is reliable enough to support a real maintenance decision. A sensor can itself be faulty or noisy, and an unusual reading does not always mean that the component is failing.

3.1 False Positive and False Negative

Situation	Meaning	Possible consequence
False positive	Healthy component is predicted as likely to fail	Unnecessary inspection, replacement, cost or delay
False negative	Faulty component is predicted as normal	Fault may remain undetected and safety may be affected

The second situation is particularly important in a safety-related system. The objective should therefore not be to maximize accuracy alone. The model must be evaluated according to the consequences of different types of errors.

3.2 Socio-Technical Nature of the Problem

Predictive maintenance is not only a machine learning problem. The output of the model becomes part of a larger human and organisational process. A maintenance engineer may use the prediction along with inspection results, maintenance history and practical knowledge. A wrong prediction can affect railway workers, passengers, train operations, maintenance schedules, costs and public confidence.

- Passengers depend on safe and reliable railway operations.
- Maintenance engineers need information that they can understand and verify.
- Railway authorities need clear responsibility when AI-assisted decisions are made.
- Operations teams may be affected by unnecessary maintenance alerts and delays.
- AI developers must account for model limitations, data quality and changing operating conditions.
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4. PROPOSED AI-BASED PREDICTIVE MAINTANENCE PROCESS

The proposed workflow is: Railway Equipment → Sensor/Inspection Data → Data Cleaning → Feature Extraction → Predictive Model → Failure Risk and Confidence → Explainability → Human Maintenance Review → Final Decision.

5. ELSI FRAMEWORK

AI-based railway predictive maintenance has ethical, legal and social implications because its predictions can influence decisions related to equipment condition and passenger safety. The following framework connects the technical system with its wider consequences.

ELSI area	Case-specific concern	Responsible AI response
Ethical	A wrong prediction may cause unnecessary maintenance or a missed fault.	Safety-focused testing, human verification and suitable thresholds.
Legal	Operational, maintenance and employee-related information may require protection.	Access control, secure storage, audit logs and applicable compliance.
Social	Passengers and workers depend on reliable decisions; repeated errors can reduce trust.	Transparent procedures, human accountability and continuous evaluation.
Accountability	It may be unclear who is responsible when an AI recommendation is wrong.	Define roles for developers, operators and maintenance personnel.
Transparency	Engineers may not understand why a component was marked high-risk.	Provide understandable explanations and document model limits.
Public safety	A missed fault can have consequences beyond the individual component.	Human oversight, conservative safety procedures and fallback mechanisms.

5.1 Stakeholders

- Passengers and the general public
- Railway maintenance engineers and technical staff
- Train operators and operations teams
- Railway authorities and decision-makers
- AI developers and data engineers
- Relevant government and regulatory authorities

5.2 Main ELSI Question

The central question is not simply whether the AI model can predict failures. It is whether the prediction can be used safely and responsibly by people who remain accountable for the final maintenance decision.

6. APPLICABLE PRINCIPLES OF RESPONSIBLE AI

RAI principle	Application in this case
Safety & reliability	Prioritize safe failure detection and test the system extensively before use.
Fairness	Check performance across equipment types, routes and operating conditions.
Transparency	Document data sources, thresholds, assumptions and limitations.
Explainability	Use SHAP or feature importance to show influential sensor features.
Accountability	Keep responsibility for final maintenance decisions clearly assigned.
Human oversight	Review high-risk and uncertain predictions before action.
Privacy	Collect and retain only information needed for the maintenance purpose.
Security	Protect sensors, records, models and predictions from manipulation.
Robustness	Test behavior with noisy, missing and unusual sensor readings.

7.EXPLAINABILITY AND INTERPRETABILITY METHODS CHOSEN

A key question in predictive maintenance is: “Why did the AI predict that this component may fail?” For a safety-related application, maintenance personnel should not be expected to blindly trust a numerical risk score.

7.1 SHAP

SHAP (SHapley Additive exPlanations) can show how individual input features contribute to a model prediction. For example, a high vibration value and an increased temperature may push the predicted failure risk upward, while a normal operating speed may push it downward. This gives an engineer a more useful view of the prediction than a single risk number.

7.2 Feature Importance

Feature importance can be used to understand which variables are generally influential in the trained model. For a sensor-based system, the result might indicate that vibration, temperature, operating time and load are among the most important features. This is useful for understanding the model at an overall level.

7.3 LIME

LIME can explain an individual prediction by approximating the model locally and identifying the features that matter most for that particular case. It can be used as a second explanation method when investigating an unusual alert.

7.4 Counterfactual Explanations

A counterfactual explanation asks what would need to change for the model's output to become different. For example, the explanation may indicate that bringing vibration and temperature readings closer to the normal range would reduce the predicted risk. This can help engineers understand the conditions associated with an alert.

7.5 Selected Primary Method

For the proposed tabular or sensor-data prototype, SHAP is selected as the primary explainability method. The intended workflow is: Sensor Data → AI Prediction → SHAP Explanation → Engineer Review → Maintenance Decision.

8. Why Explainability Matters Here

- It helps maintenance personnel investigate an alert instead of receiving only a risk score.
- It can reveal whether the model is relying on sensible sensor patterns or suspicious features.
- It supports communication between technical teams and domain experts.
- It can make model errors easier to investigate after an incident.
- It supports accountability because the reasoning behind an AI recommendation can be recorded.

9. SAFETY SECURITY AND PRIVACY

9.1 Safety

Safety is the most important concern in this case. The system should be designed so that an AI error does not automatically become a critical operational decision.

- Monitor false negatives and false positives separately.
- Use confidence thresholds and route uncertain cases to human review.
- Validate sensor readings before using them for high-impact predictions.
- Test the model under unusual and changing operating conditions.
- Maintain fallback procedures when the AI system is unavailable or unreliable.
- Periodically re-evaluate the model after deployment.

A practical rule is that a low-confidence prediction should not directly result in a critical maintenance decision. The AI should support the engineer rather than silently replacing the engineer's judgement.

9.2 Security

A connected predictive maintenance system could be affected if sensor data, maintenance records or model outputs are manipulated. Even when the model itself is accurate, compromised input data can lead to an incorrect prediction.

- Unauthorised access to monitoring systems
- Manipulation of sensor readings
- Data poisoning during model training
- Tampering with model files or prediction services
- Unauthorised changes to maintenance records
- Cyberattacks on connected monitoring infrastructure

Possible controls include authentication, role-based access, encryption, secure storage, audit logs, integrity checks and regular security testing.

9.3 Privacy

Predictive maintenance mainly concerns equipment and operational data, so privacy is less central than it would be in cases involving medical or biometric information. However, maintenance systems may still contain information related to employees, operators, locations, schedules and work records.

- Collect only data needed for the maintenance purpose.
- Restrict access according to job responsibility.
- Avoid retaining personal information unnecessarily.
- Keep appropriate records of access and data use.

10. DATA QUALITY AND ROBUSTNESS

The quality of the model depends heavily on the quality of the sensor data. Missing readings, noisy measurements, faulty sensors, changes in equipment type and changes in environmental conditions can all affect predictions.

11. ANALYSIS

11.1 False Positives and False Negatives

The two major error types have different operational consequences. A false positive can cause unnecessary inspection or replacement, increasing maintenance effort and potentially affecting schedules. A false negative means that an actual developing fault is not identified by the model. In a safety-related environment, this error needs particularly careful attention.

11.2 Data Quality

Training data may not perfectly represent real railway operations. Data can contain missing values, sensor noise, incorrect labels or limited examples of actual failures. A model trained mainly on one equipment type may also behave differently when applied to another type.

11.3 Model Bias

Bias in this case does not necessarily refer only to demographic groups. The model can behave differently across different equipment models, routes, environmental conditions, operating loads, component ages or maintenance practices.

For example, if most training examples come from a particular type of equipment, the model may learn patterns that do not transfer well to another type. Performance should therefore be checked across relevant operational groups rather than relying only on one overall accuracy value.

11.4 Human Factors

An AI system can also introduce a human-factors problem. If engineers begin to assume that the model is always correct, they may overlook evidence that conflicts with the prediction. This is sometimes described as automation bias. The system should therefore make it clear that an AI prediction is a recommendation, not a final authority.

11.5 Accountability

A responsible deployment should define who can act on an AI alert, who can override it, who maintains the model, and who investigates repeated errors. The system should keep an audit trail containing the prediction, relevant input information, explanation, human decision and subsequent outcome where appropriate.

12. PROPOSED RESPONSIBLE AI FRAMEWORK

Step	Responsible action
Sensor data	Collect condition information and validate sensor health.
AI prediction	Generate failure risk and confidence.
Explainability	Provide important features contributing to the prediction.
Risk review	Send high-risk or uncertain cases for expert review.
Human decision	Engineer decides inspection, repair or continued monitoring.
Audit	Record the prediction, explanation, decision and outcome.
Monitoring	Check model performance and retrain when required.

13. PROPOSED IMPLEMENTATION

A small proof-of-concept implementation can demonstrate the technical and Responsible AI aspects without attempting to build a complete railway maintenance platform. The implementation can use a suitable public predictive-maintenance dataset containing sensor measurements and a failure or maintenance label.

13.1 Implementation Steps

- Obtain a suitable predictive-maintenance dataset.
- Inspect the variables and understand the target label.
- Handle missing values, duplicates and obvious data-quality issues.
- Split the data into training and testing sets.
- Train a baseline classification model.
- Evaluate accuracy, precision, recall and F1-score.
- Generate a confusion matrix and specifically inspect false negatives.
- Use SHAP to explain selected predictions.
- Compare the model's predictions with actual outcomes.
- Discuss whether the model would be safe to use without human review.

13.2 Suggested Models

Model	Reason for considering it
Logistic Regression	Simple baseline and relatively easy to interpret.
Decision Tree	Produces decision rules that are easy to inspect.
Random Forest	Strong baseline for tabular data and works well with feature explanations.
XGBoost	Useful for comparing performance with a stronger boosting model.

14. EVALUATION AND RESPONSIBLE AI ANALYSIS

14.1 Technical Evaluation

The model should be evaluated using accuracy, precision, recall, F1-score and a confusion matrix. False-positive and false-negative rates should also be discussed. For predictive maintenance, accuracy alone can be misleading. A model can obtain a high overall accuracy while still missing a meaningful number of actual failures.

Area	Question to ask
Safety	How many actual failures were missed by the model?
Reliability	Does performance remain stable under different operating conditions?
Explainability	Can engineers understand important predictions?
Human oversight	Are uncertain or high-risk cases reviewed by people?
Fairness	Does performance vary across equipment types or conditions?
Security	Could sensor data or predictions be manipulated?
Privacy	Is unnecessary personal information collected?
Accountability	Can the final decision and responsible person be traced?

14.2 Benefits

- Early identification of possible equipment problems.
- Better planning of inspections and maintenance work.
- Potential reduction in unexpected failures.
- More focused use of maintenance resources.
- Additional evidence for engineers through explainability.
- Continuous monitoring when sensor infrastructure is available.

14.3 Limitations

- AI predictions can still be wrong.
- False negatives can create serious safety concerns.
- False positives can cause unnecessary maintenance and operational effort.
- Poor-quality sensor data can reduce model reliability.
- A public dataset may not represent actual Indian railway conditions.
- Historical failure labels may contain errors or omissions.
- Explainability methods do not guarantee that the model's reasoning is scientifically correct.
- The model may need retraining when equipment or operating conditions change.
- Connected monitoring systems introduce cybersecurity risks.
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14.4 Deployment Considerations

Before real deployment, the system would require extensive validation with domain experts, representative data, safety procedures, cybersecurity controls and clear operational responsibilities. The model should be monitored after deployment because data distributions and equipment conditions can change over time.

A responsible deployment should also allow engineers to challenge or override AI recommendations. This prevents a prediction from becoming an unquestioned instruction and keeps human judgement involved in safety-critical decisions.

16. CONCLUSION

AI-based predictive maintenance has the potential to support railway maintenance by identifying possible equipment problems before they become serious. However, railway maintenance is a safety-related area, so an incorrect AI prediction cannot be treated like an ordinary prediction error.

The most important distinction is between false positives and false negatives. An unnecessary maintenance alert may increase cost and operational effort, while a missed fault may create a much more serious safety concern. For this reason, the objective should not be simply to build the most accurate model. The complete decision process around the model also needs to be safe, understandable and accountable.

This case study proposes a Responsible AI approach in which AI is used as a decision-support system rather than an independent decision-maker. Sensor data can be analysed by a machine learning model, while SHAP and other explainability techniques can help maintenance engineers understand the reasons behind predictions.

Human oversight is especially important for high-risk and uncertain cases. The system should also be evaluated for reliability, robustness, security, data quality and changes in operating conditions. The final maintenance decision should remain with an authorised human expert who can consider information that may not be available to the model.

AI predicts the possible failure → AI explains the prediction → Human expert makes the final decision.

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